

UCE824 :GEOTECHNICES FOR UNDERGROUND STRUCTURES				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: To enable students in visualizing and critically analysing the behavior of underground structures with reference to various supporting systems under different loading conditions due to induced earth pressure				
Excavation methods and lateral supporting system: Introduction, excavation methods and lateral supporting systems, retaining walls, strutting systems, factors influencing on the selection of the retaining strut system, case history. Types of sheet piles, principal advantages of sheet piles, analysis of cantilever wall in sands, and clays, Anchored bulk head stability, Free earth support method, fixed earth support method, Types and Design of anchor systems				
Coffer Dams: Types of cofferdams, relative merits and their advantages as compared to other types, comparison between circular and diaphragm types, failure modes of cells, stability analysis of cofferdams				
Open cuts: necessity of bracing and strutting in open cuts, pressure distribution diagram under various cases, deep open cut in loose and dense sands, deep open cut in normally loaded and stiff clays. Stability Analysis of open cuts				
Excavation and Protection of Adjacent Structures: Introduction, protection of building using the behaviour of excavation induced deformation, building protection by auxiliary methods, construction defects and remedial measures, building rectification methods				
Dewatering: Methods of Dewatering, Ditches and sumps, well point system, deep well drainage installation, vacuum method Bleeder wells, sand drain installation, electro-osmosis. Design of dewatering system for deep excavations				
Trenchless Technology: Introduction, methods like Horizontal Auger Boring, Horizontal Directional Boring, Directional Drilling, Micro-tunneling, Pilot Tube Micro-tunneling, Pipe Jacking, Moling, Hydroexcavation, Guided Drilling, Jack and Bore				
Tunnelling: Introduction , types of tunnels, , Various Construction Methods of tunnelling like Drill and blast, TBM, NATM, SCL. Tunnel lining system Primary and Secondary support system based on rock mass classification, RMR system and Q- system. Ground improvement techniques in Tunnel				
Course Learning Objectives (CLO) Upon completion of this course, the students will be able to: 1. Analyze the underground structures with reference to various supporting systems that needs for underground construction 2. Design the underground structures 3. Develop an ability to protect the adjacent building due to underground construction.				
Reference Books:				

1. Deep Excavation Theory and Practice by Chang – Yu Ou, Taylor & Francis Group, London, UK, 2006.
2. Gopal Ranjan& A.S.R. Rao, Basic and Applied Soil mechanics, New Age Publisher, New Delhi (2016)
3. Foundation Engineering by Peck, R. B., Hanson, W.E., and Thornburn, T.H., John Wiley & Sons,
4. Foundation Analysis and Design by Bowles, J. E. 4th Ed. McGraw – Hill Book Company, New York, USA, 2009

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	30
2.	EST	45
3.	Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations)	25